

CropMitra

Agricultural Decision Support System

Current Implementation & Technical Project Report

Project Focus: Multi-Criteria Crop & Fertilizer Recommendation

Core Technologies: React 18, FastAPI, Python, scikit-learn, Random Forest, SQLite

Primary Datasets: CropDataset-Enhanced.csv, soil info.csv, fertilizer.csv,
crop_rotation_matrix.csv

Author / Engineering: CropMitra Core Development Team

Audit Date: September 2026 | Version 2.0.0 (Production Verified)

Confidentiality & Academic Notice: This report represents a factual audit of the actual codebase, datasets, and tested software artifacts present in the CropMitra repository. All performance metrics and claims are verified against automated test suites.

1. Executive Summary

CropMitra is an intelligent, multi-criteria agricultural decision-support web application designed to assist Indian smallholder farmers, agronomists, and agricultural extension officers in making data-driven crop selection and fertilization decisions.

Selecting the optimal crop for a given plot is complex. A farmer must jointly weigh geographical adaptation, field-level soil nutrient profiles, seasonal water availability, preceding crop sequence history (crop rotation), and targeted fertilization needs. CropMitra addresses this multi-dimensional challenge through an integrated, layered software architecture:

- **Frontend:** Responsive SPA built with React 18 and Vite, featuring dynamic location selectors, interactive field parameters, explainable recommendation cards, and print views.
- **Backend:** High-performance RESTful API built on Python 3.14 and FastAPI with strict Pydantic validation.
- **ML & Decision Engine:** Hard location filtering from 730 district records, 99.32% accurate Random Forest model, Gaussian environmental similarity envelopes, and an in-memory 2,601-pair directional rotation matrix.
- **Database:** Local SQLite database managed via SQLAlchemy ORM for recommendation session persistence.
- **Fertilizer Engine:** Target-crop and soil-specific NPK formulation guidance with stage-wise split dosage advice.

2. Problem Statement

Indian agriculture faces persistent structural productivity barriers due to lack of localized decision support:

- **Location Mismatch:** Sowing crops unadapted to district agro-climatic zones leads to widespread crop failures.
- **Imbalanced Fertilizer Usage:** Heavy overuse of synthetic Urea without balanced P and K degrades soil microbial health.
- **Continuous Monoculture:** Cultivating identical crops repeatedly accumulates soil-borne pests, wilt, and root-knot nematodes.
- **Water Incompatibility:** Misaligning high-water crops with rain-fed dry plots causes drought stress and economic losses.

CropMitra provides an accessible, transparent, and explainable decision-support tool to mitigate these issues.

3. Implemented Project Objectives

1. Provide multi-candidate crop recommendations with transparent suitability tiers and scores.
2. Incorporate real field parameters: Soil Type, Water Availability, and Previous Crop.
3. Deploy a trained Random Forest classifier to predict suitability envelopes across environmental metrics.
4. Enforce strict geographic location authority via historical district crop registries.
5. Implement directional crop rotation evaluation across 2,601 crop pairs based on nitrogen fixation and pest break rules.
6. Provide explainable plain-language rationales for every recommendation.
7. Deliver precise fertilizer formulations, dosages (kg/acre), and split-application guidance.

- 8. Persist recommendation sessions in an SQLite database.
- 9. Deliver a mobile-responsive, print-ready React UI with dark and light earthy themes.

4. System Architecture

The application follows a clean 3-tier decoupled architecture:

Client Tier: React 18 + Vite SPA (Forms, Interactive Selectors, Result Banners, Print View)

API & Validation Tier: FastAPI (REST Endpoints, Pydantic Request Models, CORS)

Decision & ML Tier: Location Service + Random Forest Model + Crop Rotation Service + Fertilizer Service

Persistence Tier: SQLAlchemy ORM + SQLite (RecommendationRecord Table)

5. Technology Stack

Technology	Layer / Purpose	Actual Usage in Project
React 18	Frontend Framework	SPA UI, state management, form validation
Vite 5	Frontend Build Tool	Fast HMR and production bundle compilation
Vanilla CSS	UI Styling System	Custom variables, dark/light themes, print CSS
Python 3.14	Backend Runtime	FastAPI server, ML pipeline, dataset generation
FastAPI 0.115	REST API Framework	High-speed asynchronous routing and validation
Pydantic 2.10	Schema Validation	Strict input/output data typing
scikit-learn 1.6	Machine Learning	RandomForestClassifier for continuous condition prediction
pandas & NumPy	Data & Math Engine	Tabular indexing, Gaussian distance calculation
joblib	Model Serialization	Serializing & loading .joblib model files
SQLAlchemy 2.0	Database ORM	Schema modeling & SQLite query abstraction
SQLite 3	Local Database	Local persistent logging in cropmitra.db
pytest 9.0	Automated Testing	Unit, integration, and regression test suites

6. Datasets Audit & Provenance

Dataset File	Rows	Cols	Role & Integration Status
soil_info.csv	2,200	8	ML Random Forest training for N,P,K,Temp,Humidity,pH,Rainfall
fertilizer.csv	8,000	9	Baseline fertilizer classifier training data
CropDataset-Enhanced.csv	730	23	Authoritative geographical district-to-crop registry (Hard Filter)
crop_rotation_matrix.csv	2,601	14	Runtime in-memory directional sequence evaluation matrix
crop_rotation_dataset.xlsx	2,601	14	Multi-sheet formatted Excel reference workbook (4 sheets)

7. Machine Learning & Multi-Criteria Scoring

The crop recommendation engine uses a normalized, multi-criteria decision formula ensuring mathematical rigor and zero fabricated multipliers:

Normalized Multi-Factor Formula:

```
base_score = 0.50 * ml_score + 0.30 * condition_score + 0.20 * rotation_score
final_score = round(min(9.9, max(2.0, base_score * loc_mult * 10.0)), 1)
```

- **ml_score [0,1]:** Random Forest prediction probability blended with Gaussian condition similarity.
- **condition_score [0,1]:** Soil type compatibility and water availability constraints.
- **rotation_score [0,1]:** Directional crop sequence rating from the in-memory rotation matrix.
- **loc_mult:** 1.05 for exact district-level match, 1.00 for state fallback.
- **Weights:** 0.50 + 0.30 + 0.20 = 1.0000 strictly.

8. Proof-of-Impact Verification (Maharashtra -> Kolhapur)

Auditing all candidate crops under identical field conditions (Medium Water, Loamy Soil):

Candidate Crop	Prev: Wheat	Prev: Sugarcane	Prev: Soybean	Rot Impact / Shift
Soybean	9.4 (Rot: 0.85, #1)	9.4 (Rot: 0.85, #1)	8.3 (Avoid, #4)	Monoculture penalty under Soybean
Groundnut	9.1 (Rot: 0.85, #2)	9.1 (Rot: 0.85, #2)	8.6 (Rot: 0.60, #3)	Stable high performance
Sugarcane	8.9 (Rot: 0.75, #3)	8.1 (Avoid, #4)	8.9 (Rot: 0.75, #2)	Demoted out of top 3 under Sugarcane
Jowar (Sorghum)	8.5 (Rot: 0.60, #4)	8.6 (Rot: 0.65, #3)	9.0 (Rot: 0.85, #1)	Promoted to Rank #1 under Soybean
Rice	6.5 (Rot: 0.60, #5)	6.6 (Rot: 0.65, #5)	7.0 (Rot: 0.85, #5)	Lowest ML suitability in district

9. Backend REST API Endpoints

Method	Endpoint	Request / Params	Purpose
GET	/api/health	None	Server health check and model loading verification
GET	/api/locations/states	None	Returns 36 Indian states and union territories
GET	/api/locations/districts	state: str	Returns districts registered for selected state
GET	/api/locations/crops	state: str, district: str	Returns supported crops for selected location
GET	/api/crops	None	Returns 51 canonical crops for general dropdowns
POST	/api/recommend	CropRecommendRequest	Full 5-stage crop recommendation pipeline
POST	/api/fertilizer	FertilizerRequest	NPK dosage and split-application guidelines
POST	/api/recommendations/save	SaveRecommendationRequest	Persists session record into SQLite database
GET	/api/recommendations	limit: int = 20	Retrieves historical recommendation records

10. Automated Testing & Verification

Test Suite	Tests	Status	Verified Capabilities
backend/test_backend.py	8 Tests	PASSED	API health, location registry, Akola/Kolhapur/Latur filters, DB
backend/test_rotation.py	11 Tests	PASSED	Sugarcane->Soybean, Wheat->Wheat, unknown fallback, Proof of Impact
ML Evaluation	2,200	PASSED	99.32% test accuracy on 20% stratified test set
Frontend Production Build	1,886	PASSED	Vite build compiled in 4.36s with 0 errors
Determinism Test	5 Runs	PASSED	100% identical rankings and scores across identical requests

11. Actual Implementation Status Audit

Feature / Component	Implementation Status	Direct Evidence in Repository
React Frontend SPA	IMPLEMENTED	frontend/src/App.jsx, pages/, index.css
FastAPI REST Backend	IMPLEMENTED	backend/app/main.py, api/routes.py
Random Forest Crop Model	IMPLEMENTED	backend/ml/models/crop_model.joblib (99.32%)
Location District Registry	IMPLEMENTED	backend/data/CropDataset-Enhanced.csv (730 rows)
Hard Location Crop Filtering	IMPLEMENTED	backend/app/services/location_service.py
Deterministic Crop Rotation	IMPLEMENTED	backend/app/services/rotation_service.py (2,601 pairs)
Multi-Sheet Excel Workbook	IMPLEMENTED	backend/data/crop_rotation_dataset.xlsx (4 sheets)
Fertilizer Recommendation System	IMPLEMENTED	backend/app/services/fertilizer_service.py
SQLite Database & SQLAlchemy	IMPLEMENTED	backend/app/database/models.py, cropmitra.db
Recommendation Persistence & History	IMPLEMENTED	POST /api/recommendations/save, GET /api/recommendations
Print-Ready Advisory Stylesheet	IMPLEMENTED	frontend/src/index.css (@media print)
Dark & Light Earthy Themes	IMPLEMENTED	frontend/src/index.css (data-theme toggle)
Mobile Responsive UI	IMPLEMENTED	frontend/src/index.css (Fluid media queries)
Live IoT Soil Sensor Hardware	PLANNED / NOT IMPLEMENTED	Planned for future hardware expansion
Real-Time Satellite Weather API	PLANNED / NOT IMPLEMENTED	Planned for future API integration
Regional Multilingual Localization	PLANNED / NOT IMPLEMENTED	Planned for Indian regional languages

12. Conclusion & Presentation Summary

Executive Conclusion: CropMitra is a fully functional, mathematically normalized, and scientifically grounded agricultural decision-support web platform. By integrating geographical district evidence, machine learning condition predictions, physical soil/water constraints, and directional crop rotation rules, CropMitra delivers robust, explainable, and reproducible crop and fertilizer advisories.

Viva Summary: CropMitra does not replace lab soil testing or extension officers; rather, it provides an intuitive digital decision-support tool bridging complex university agricultural standards with real-world smallholder farm planning.